

CISCO CERTIFIED CYBEROPS ASSOCIATE (CBROPS 200-201)

Lab 10.2.7: Using Wireshark to Examine a UDP DNS Capture

Transport Layer Analysis, UDP Header Mechanics, DNS Query/Response Dissection & Layer 2/3 Address Resolution

COURSE MODULE	PROTOCOLS ANALYZED	ANALYSIS TOOL	ENVIRONMENT
Module 10: Network Services	UDP (RFC 768) & DNS (RFC 1035)	Wireshark & Linux Network CLI	CyberOps Workstation VM (enp0s3)

Objectives

- **Part 1: Record IP Configuration Information of the PC** — Document MAC address, IPv4 address, DNS nameserver configuration, and default gateway routing table.
- **Part 2: Use Wireshark to Capture DNS Queries and Responses** — Capture live traffic on interface `enp0s3` while resolving `www.google.com`.
- **Part 3: Analyze Captured DNS or UDP Packets** — Dissect the 8-byte UDP transport header, inspect DNS query/response structures, and evaluate Layer 2 (MAC) vs. Layer 3 (IP) routing dynamics.
- **Reflection & Architectural Evaluation** — Assess the performance, scalability, and overhead advantages of UDP vs. TCP for DNS transport.

Background / Scenario

The User Datagram Protocol (UDP) is an unacknowledged, connectionless Transport Layer protocol operating at OSI Layer 4. Unlike TCP, UDP introduces minimal protocol overhead (a fixed 8-byte header) and requires no connection setup handshake or state tracking. Because DNS lookups require rapid, lightweight transactions, UDP port 53 is the primary protocol used for domain name queries.

In this lab, network parameters of the CyberOps Workstation VM are recorded, followed by packet capture and forensic dissection of a live DNS resolution sequence in Wireshark.

Part 1: Record the IP Configuration Information of the PC

Executing network diagnostic commands from the CyberOps Workstation CLI reveals the host's networking baseline:

1. Inspect Physical Interface and MAC Address:

```
[analyst@secOps ~]$ ip link
```

```
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:da:28:83 brd ff:ff:ff:ff:ff:ff
```

2. Inspect Assigned IPv4 Address and Subnet Mask:

```
[analyst@secOps ~]$ ip addr
```

```
2: enp0s3: ...
    inet 192.168.8.10/24 brd 192.168.8.255 scope global dynamic enp0s3
```

3. Inspect Configured DNS Resolvers:

```
[analyst@secOps ~]$ cat /etc/resolv.conf
```

```
nameserver 8.8.4.4
nameserver 8.8.8.8
nameserver 209.165.200.235
```

4. Display Kernel IP Routing Table & Default Gateway:

```
[analyst@secOps ~]$ netstat -rn
```

```
Kernel IP routing table
```

Destination	Gateway	Genmask	Flags	MSS Window	irtt	Iface
0.0.0.0	192.168.8.1	0.0.0.0	UG	0 0	0	enp0s3
192.168.8.0	0.0.0.0	255.255.255.0	U	0 0	0	enp0s3

NETWORK PARAMETER	RECORDED HOST VALUE	DIAGNOSTIC SOURCE COMMAND	ROLE IN DNS QUERY TRANSMISSION
Host MAC Address	08:00:27:da:28:83	ip link / ifconfig	Layer 2 Source Hardware Address for all outgoing frames.
Host IPv4 Address	192.168.8.10	ip addr	Layer 3 Source IP in query packets.
Subnet Mask	255.255.255.0 (/24)	ip addr	Defines local broadcast domain boundary.
Default Gateway IP	192.168.8.1	netstat -rn / ip route	Next-hop router interface for off-subnet packets.
DNS Server IP	8.8.4.4 (Google Public DNS)	cat /etc/resolv.conf	Layer 3 Destination IP for domain queries.

Part 2: Use Wireshark to Capture DNS Queries and Responses

Wireshark is launched on interface `enp0s3`. A web browser navigates to `www.google.com` (or alternatively `ping -c 2 www.google.com` is run in terminal). Once Google's IP is resolved, capture is stopped and filtered with `dns`.

Part 3: Analyze Captured DNS and UDP Packets

Step 1 & 2: Dissect the DNS Query Packet (Frame 429)

STEP 2.B — LAYER 2 SOURCE MAC VERIFICATION

Is the source MAC address the same as the one recorded from Part 1 for the VM?

ANALYST FINDING:

Yes. The Source MAC address displayed in the Ethernet II frame (`08:00:27:da:28:83`) is identical to the physical MAC address recorded for the VM's active network interface (`enp0s3`) in Part 1.

STEP 2.C — LAYER 2 VS. LAYER 3 ADDRESSING IDENTIFICATION

Can you identify the IP and MAC addresses for the source and destinations of this packet?

ADDRESSING MATRIX:

DEVICE ROLE	IPV4 ADDRESS (LAYER 3)	MAC ADDRESS (LAYER 2)	ROUTING / ENCAPSULATION EXPLANATION
Source Workstation (VM)	192.168.8.10	08:00:27:da:28:83	Originating endpoint that generated the DNS lookup.
Destination (DNS Server / Gateway)	8.8.4.4 (DNS Server)	50:c7:bf:dc:3a:41 (Default Gateway)	Crucial Concept: The Destination IP is the ultimate endpoint (DNS server 8.8.4.4), but the Destination MAC is the local Default Gateway router , which forwards the frame off the local LAN.

UDP Transport Header Anatomy (RFC 768)

The UDP segment header contains exactly **4 fields (8 bytes total)**, with each field occupying exactly 16 bits (2 bytes):

FIELD NAME	FIELD SIZE	CAPTURED PACKET VALUE	TECHNICAL DESCRIPTION
Source Port	16 bits (2 bytes)	58029	Dynamically allocated client ephemeral port used to track the response.
Destination Port	16 bits (2 bytes)	53	Well-known port reserved for DNS servers listening for incoming queries.
Length	16 bits (2 bytes)	40 bytes	Total length of UDP segment: 8 bytes (header) + 32 bytes (DNS payload).
Checksum	16 bits (2 bytes)	0x7d2e [verified]	Validates segment data integrity across network transit.

STEP 2.D — ADDRESS CORRELATION VERIFICATION

1. Is the source IP address the same as the local PC's IP address you recorded in Part 1?
2. Is the destination IP address the same as the default gateway noted in Part 1?

ANALYST FINDINGS:

- **1. Source IP Correlation: Yes.** The source IP (192.168.8.10) matches the VM's assigned IP address recorded in Part 1.
- **2. Destination IP Correlation: No (in general).** The destination IP address (8.8.4.4) is the configured public DNS recursive resolver from /etc/resolv.conf , whereas the default gateway is 192.168.8.1 . (Note: While the destination MAC address matches the Default Gateway router, the destination IP is the upstream DNS server).

Step 3: Dissect the DNS Response Packet (Frame 488)

STEP 3.B — ETHERNET II ADDRESSING ROLES IN RESPONSE

In the Ethernet II frame for the DNS response, what device is the source MAC address and what device is the destination MAC address?

DISSECTION FINDING:

- **Source MAC Address:** The Default Gateway router's LAN interface MAC address (`50:c7:bf:dc:3a:41`).
- **Destination MAC Address:** The Workstation VM's virtual NIC MAC address (`08:00:27:da:28:83`).

STEP 3.C — NETWORK & TRANSPORT LAYER ROLE REVERSAL

1. What is the destination IP address?
2. What is the source IP address?
3. What happened to the roles of source and destination for the VM and default gateway?

SOC ANALYTICAL ANSWERS:

- **1. Destination IP Address:** `192.168.8.10` (CyberOps Workstation VM).
- **2. Source IP Address:** `8.8.4.4` (DNS Server).
- **3. Role Inversion Analysis:** The roles of source and destination across **all OSI layers (Layer 2 MAC, Layer 3 IP, and Layer 4 Port numbers)** are **completely reversed/inverted**. The DNS server replies back to the exact ephemeral port (`58029`) and host IP that initiated the query, allowing the host OS to route the response to the requesting application.

Reflection Question & Architectural Evaluation

REFLECTION QUESTION — TRANSPORT LAYER PROTOCOL SELECTION (UDP VS. TCP FOR DNS)

What are the benefits of using UDP instead of TCP as a transport protocol for DNS?

COMPREHENSIVE ENGINEERING & CYBEROPS EVALUATION:

1. Minimal Protocol Overhead & Reduced Bandwidth:

A UDP header is fixed at only **8 bytes**, whereas a TCP header consumes a minimum of **20 bytes** (up to 60 bytes with options). For short query-response cycles, UDP significantly reduces bandwidth consumption across high-volume networks.

2. Zero Handshake Latency (No 3-Way Handshake):

UDP is connectionless. A DNS client sends its query datagram immediately without establishing a connection. In contrast, TCP requires a 3-way handshake (`SYN → SYN-ACK → ACK`) and teardown (`FIN/RST`), which introduces at least 1.5 to 2 Round Trip Times (RTT) of added latency before query data can be transmitted.

3. Stateless Scalability on DNS Servers:

Root and recursive DNS servers handle millions of queries per second. With UDP, the server processes datagrams statelessly without allocating operating system memory buffers or maintaining Transmission Control Blocks (TCBs) for sequence numbers, window sizing, and retransmission timers.

4. Application-Layer Reliability Handling:

Because DNS queries are idempotent (repeating a query produces the same result without side effects), DNS client applications handle retransmissions directly with a simple retry timer, rendering heavy TCP retransmission mechanisms unnecessary for small standard lookups.

When DNS Uses TCP Port 53: TCP is used when DNS response payloads exceed 512 bytes (without EDNS0), for **DNS Zone Transfers (AXFR / IXFR)** between primary and secondary nameservers, and for **DNS over TLS (DoT - RFC 7858)** on port 853.

CyberOps Protocol Analysis Summary

- **Layer 2 vs 3 Dynamics** IP routing ensures packets reach remote destination IPs (`8.8.4.4`), while Ethernet MAC addressing directs frames hop-by-hop to local default gateways (`192.168.8.1`).
- **UDP Simplicity** UDP provides fast, lightweight 8-byte transport without connection overhead for time-sensitive application lookups.
- **Security Implication** Because UDP is connectionless and unauthenticated, standard DNS is vulnerable to **IP spoofing** and **DNS amplification DDoS attacks**, mandating defensive measures like **Response Rate Limiting (RRL)** and **DNSSEC**.